Relaxation and Odd-Even Effect in Fermi Liquids

Kittiphat Pongarunotai, Puripat Thumbanthu

Outline

1 Introduction

1.1 Boltzmann equation

Boltzmann equation

$$\left(\frac{\partial}{\partial t} + \mathbf{v} \cdot \frac{\partial}{\partial \mathbf{r}} + \mathbf{F} \cdot \frac{\partial}{\partial \hbar \mathbf{p}}\right) f(t, \mathbf{x}, \mathbf{p}) = \hat{\mathcal{J}}[f(t, \mathbf{x}, \mathbf{p})] \tag{1}$$

$f(t, \mathbf{x}, \mathbf{p})$: Chance of finding a particle at $\mathbf{x}$ with momentum $\mathbf{p}$ at time t.

$\frac{\partial f}{\partial t}$: Local time evolution

$\mathbf{v} \cdot \frac{\partial f}{\partial \mathbf{r}}$: Streaming/advection term

$\mathbf{F} \cdot \frac{\partial f}{\partial \hbar \mathbf{p}}$: Force/acceleration term due to external fields

$\hat{\mathcal{J}}[f]$: Collision integral - effect of collisions on distribution function

1.2 Fermi Dirac distribution

Fermi-Dirac distribution

$$f_0(\mathbf{p}) = \frac{1}{\exp[\beta(\varepsilon(\mathbf{p}) - \mu)] + 1} \tag{2}$$

- $\beta = 1/k_B T$ is the inverse temperature
- $\varepsilon(\mathbf{p}) = \dfrac{\hbar^2 \mathbf{p}^2}{2m}$ is the energy of particles with momentum $\mathbf{p}$
- μ is the chemical potential

The Fermi-Dirac distribution is in **equilibrium**.

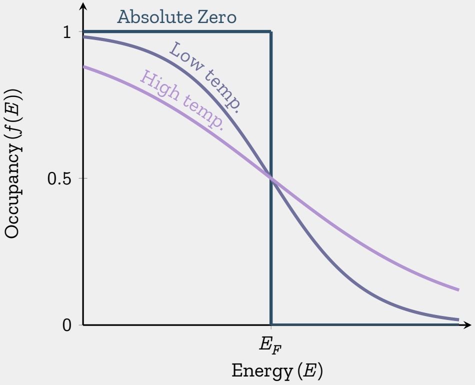

Figure 1: Fermi-Dirac distribution at different temperatures

- At $T = 0$, all states below the Fermi energy are filled, all states above are empty.
- As T increases, the distribution smooths out.

k-space

- Analogous to momentum space
- Each point $\mathbf{k}$ in k-space corresponds to a wave $e^{i\mathbf{k}}$
- **Fourier transform** converts between k-space to real space

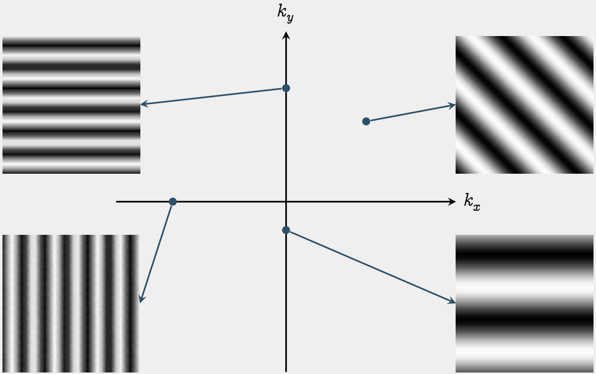

Figure 2: Representative waves of k-space

- Distance from origin $\propto$ Wave frequency
- Direction on k-space $\implies$ Wave direction

Fermi surface

- Surface in k-space that separates *occupied electron states* from *unoccupied* electron states
- Determined by periodicity and symmetry of crystal lattice.
- Changes with temperature.

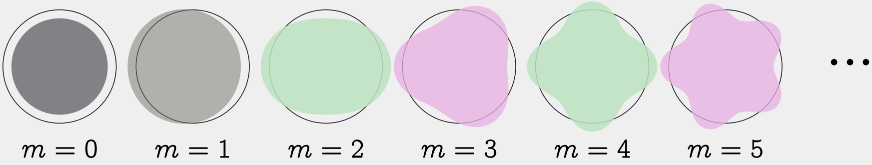

Figure 3: CIRCULAR HARMONICAL PERTURBATIONS OF THE FERMI SURFACE

Perturb the Fermi surface by external fields. For smooth perturbations, the perturbation can be expanded in harmonics basis: $e^{im\theta}$.

2 Recovering equations

2.1 Linearization of the collision integral

Ideas

- Consider **only perturbation near Fermi's equilibrium**

$$f(t, \mathbf{p}) = f_0(\mathbf{p}) + \delta f(t, \mathbf{p})$$

- Transform the integrals into a linear operator

Preparation

Fermi equilibrium: $f_0(\mathbf{p}) = \dfrac{1}{\exp[\beta(\varepsilon(\mathbf{p}) - \mu)] + 1}$

$$\frac{\partial f_0}{\partial \varepsilon} = -\frac{1}{T} f_0(1 - f_0) \tag{3}$$

$$\frac{\partial f_0}{\partial \mu} = \frac{1}{T} f_0(1 - f_0) \tag{4}$$

$$\frac{\partial f_0}{\partial \varepsilon} = -\frac{\partial f_0}{\partial \mu} \tag{5}$$

$$f(t,\mathbf{p}) = f_0(\mathbf{p}) + \delta f(t,\mathbf{p}) \qquad \text{Perturbation} \tag{6}$$

$$\approx f_0(\mathbf{p}) + \frac{\partial f_0}{\partial \mu}\,\delta\mu \qquad \text{Taylor series. Fix } \varepsilon \tag{7}$$

$$\therefore \delta f(t,\mathbf{p}) = \frac{\partial f_0}{\partial \mu}\,\delta\mu \tag{8}$$

$$f(t, \mathbf{p}) = f_0(\mathbf{p}) + \delta f(t, \mathbf{p}) \quad \text{Perturbation} \quad (6)$$

$$\approx f_0(\mathbf{p}) + \frac{\partial f_0}{\partial \mu} \delta\mu \quad \text{Taylor series. Fix } \varepsilon \quad (7)$$

$$\therefore \delta f(t, \mathbf{p}) = \frac{\partial f_0}{\partial \mu} \delta\mu \quad (8)$$

Perturbation of interest: $\delta\mu = T\psi$ (temperature dependent change in chemical potential)

$$\delta f(t, \mathbf{p}) = \frac{\partial f_0}{\partial \mu} T\psi \quad (9)$$

$$= \left(-\frac{1}{T} f_0(1 - f_0)\right) \times T\psi \quad (10)$$

$$= f_0(1 - f_0)\psi \quad (11)$$

Collision integral

$$\hat{\mathcal{J}}[f] = -\int \dots \int \frac{1}{(2\pi)^6}\, \mathrm{d}\mathbf{p}_2\, \mathrm{d}\mathbf{p}_1'\, \mathrm{d}\mathbf{p}_2' W(\mathbf{p}_1', \mathbf{p}_2' | \mathbf{p}_1, \mathbf{p}_2)\mathcal{B}, \tag{12}$$

$$\mathcal{B} = f_1 f_2 (1 - f_{1'})(1 - f_{2'}) - f_{1'} f_{2'} (1 - f_1)(1 - f_2), \tag{13}$$

$$f_i = f(\mathbf{p}_i), \quad f_{0,i} = f_0(\mathbf{p}_i). \tag{14}$$

Linearization

Taylor expansion of $\mathcal{B}$ at f_0:

$$\mathcal{B} = \mathcal{B}_{f_0} + \mathcal{B}_{\text{lin}} + \mathcal{B}_{\text{non-lin}} \tag{15}$$

The linear part is:

$$\mathcal{B}_{\text{lin}} = \left(\frac{\partial \mathcal{B}}{\partial f_1}\right)_{f_0} \delta f_1 + \left(\frac{\partial \mathcal{B}}{\partial f_2}\right)_{f_0} \delta f_2 + \left(\frac{\partial \mathcal{B}}{\partial f_1'}\right)_{f_0} \delta f_{1'} + \left(\frac{\partial \mathcal{B}}{\partial f_{2'}}\right)_{f_0} \delta f_{2'} \tag{16}$$

$$\left(\frac{\partial\mathcal{B}}{\partial f_1}\right)_{f_0} = f_{0,2}(1-f_{0,1'})(1-f_{0,2'}) + f_{0,1'}f_{0,2'}(1-f_{0,2}), \tag{17}$$

$$\left(\frac{\partial\mathcal{B}}{\partial f_2}\right)_{f_0} = f_{0,1}(1-f_{0,1'})(1-f_{0,2'}) + f_{0,1}f_{0,2}(1-f_{0,1}), \tag{18}$$

$$\left(\frac{\partial\mathcal{B}}{\partial f_{1'}}\right)_{f_0} = -f_{0,1}f_{0,2}(1-f_{0,2'}) - f_{0,2'}(1-f_{0,1})(1-f_{0,2}), \tag{19}$$

$$\left(\frac{\partial\mathcal{B}}{\partial f_{2'}}\right)_{f_0} = -f_{0,1}f_{0,2}(1-f_{0,1'}) - f_{0,1'}(1-f_{0,1})(1-f_{0,2}). \tag{20}$$

Substitute $\delta f_i = f_{0,i}(1-f_{0,i})\psi_i$. E.g.,

$$\left(\frac{\partial \mathcal{B}}{\partial f_1}\right)_{f_0} \delta f_1 = \Big[f_{0,2}(1-f_{0,1'})(1-f_{0,2'}) + f_{0,1'}f_{0,2'}(1-f_{0,2})\Big] \times f_{0,1}(1-f_{0,1})\psi_1 \tag{21}$$

$$= \Big[f_{0,1}f_{0,2}(1-f_{0,1'})(1-f_{0,2'})(1-f_{0,1}) + f_{0,1}f_{0,1'}f_{0,2'}(1-f_{0,1})(1-f_{0,2})\Big]\psi_1$$

$$= \Big(f_{0,1}f_{0,2}(1-f_{0,1'})(1-f_{0,2'})\Big)\Big[(1-f_{0,1}) + f_{0,1}\Big]\psi_1 \tag{22}$$

$$= \Big(f_{0,1}f_{0,2}(1-f_{0,1'})(1-f_{0,2'})\Big)\psi_1 \tag{23}$$

Define $F_{121'2'} = f_{0,1}f_{0,2}(1-f_{0,1'})(1-f_{0,2'}) = f_{0,1'}f_{0,2'}(1-f_{0,1})(1-f_{0,2})$. So that

$$\left(\frac{\partial \mathcal{B}}{\partial f_1}\right)_{f_0} \delta f_1 = F_{121'2'}\psi_1 \tag{24}$$

$$\left(\frac{\partial \mathcal{B}}{\partial f_i}\right)_{f_0} \delta f_i = F_{121'2'}\psi_i \tag{25}$$

Hence,

$$\mathcal{B}_{\text{lin}} = \left(\frac{\partial \mathcal{B}}{\partial f_1}\right)_{f_0} \delta f_1 + \left(\frac{\partial \mathcal{B}}{\partial f_2}\right)_{f_0} \delta f_2 + \left(\frac{\partial \mathcal{B}}{\partial f_1'}\right)_{f_0} \delta f_{1'} + \left(\frac{\partial \mathcal{B}}{\partial f_{2'}}\right)_{f_0} \delta f_{2'} \tag{26}$$

$$= F_{121'2'}\left[\psi_1 + \psi_2 - \psi_{1'} - \psi_{2'}\right] \tag{27}$$

$$= F_{121'2'}\left[\psi(\mathbf{p}_1) + \psi(\mathbf{p}_2) - \psi(\mathbf{p}_1') - \psi(\mathbf{p}_2')\right] \tag{28}$$

$$\hat{\mathcal{J}}[f] = -\frac{1}{(2\pi)^6}\int\dots\int \mathrm{d}\mathbf{p}_2\,\mathrm{d}\mathbf{p}_1\,\mathrm{d}\mathbf{p}_2' W(\mathbf{p}_1',\mathbf{p}_2'|\mathbf{p}_1,\mathbf{p}_2)\mathcal{B} \quad (29)$$

$$= -\frac{1}{(2\pi)^6}\int\dots\int \mathrm{d}\mathbf{p}_2\,\mathrm{d}\mathbf{p}_1\,\mathrm{d}\mathbf{p}_2' W(\mathbf{p}_1',\mathbf{p}_2'|\mathbf{p}_1,\mathbf{p}_2)(\mathcal{B}_{f_0} + \mathcal{B}_{\text{lin}} + \mathcal{B}_{\text{non-lin}}) \quad (30)$$

- Collision integral vanishes at equilibrium: $\hat{\mathcal{J}}[f_0] = 0$
- Throw out $\mathcal{B}_{\text{non-lin}}$

$$\hat{\mathcal{J}}[f] = -\frac{1}{(2\pi)^6}\int\dots\int \mathrm{d}\mathbf{p}_2\,\mathrm{d}\mathbf{p}_1\,\mathrm{d}\mathbf{p}_2' \times W(\mathbf{p}_1',\mathbf{p}_2'|\mathbf{p}_1,\mathbf{p}_2)F_{121'2'}[\psi(\mathbf{p}_1)+\psi(\mathbf{p}_2)-\psi(\mathbf{p}_1')-\psi(\mathbf{p}_2')] \quad (31)$$

Expression of $F_{121'2'}$

- For convenience, write $F_{121'2'}$ as **total momentum** (P) and **center of momentum** (Q)

$$\mathrm{P} = \mathbf{p}_1 + \mathbf{p}_2, \quad \mathrm{Q} = \frac{1}{2}\mathbf{p}_1 - \frac{1}{2}\mathbf{p}_2. \tag{32}$$

$$\mathbf{p}_1 = \frac{1}{2}\mathbf{P} - \mathbf{Q}, \tag{33}$$

$$\mathbf{p}_2 = \frac{1}{2}\mathbf{P} + \mathbf{Q}, \tag{34}$$

$$\mathbf{P}\cdot\mathbf{Q} = \frac{1}{2}\left(p_1^2 - p_2^2\right). \tag{35}$$

$$\varepsilon(\mathbf{p}_1) = \frac{\hbar^2 p_1^2}{2m^*} = \frac{\hbar^2}{2m^*}\left(\frac{1}{4}P^2 + Q^2 - \mathbf{P}\cdot\mathbf{Q}\right), \tag{36}$$

$$\varepsilon(\mathbf{p}_2) = \frac{\hbar^2 p_2^2}{2m^*} = \frac{\hbar^2}{2m^*}\left(\frac{1}{4}P^2 + Q^2 + \mathbf{P}\cdot\mathbf{Q}\right). \tag{37}$$

Define[1]

$$\xi = \frac{\hbar^2\beta}{2m^*}, \quad \text{and} \quad X = \beta\left(\frac{\hbar^2 P^2}{8m^*} + \frac{\hbar^2 Q^2}{2m^*} - \mu\right) \tag{38}$$

Then,

$$\beta\left(\varepsilon(\mathbf{p}_1) - \mu\right) = \beta\left(\frac{\hbar^2}{8m^*}P^2 + \frac{\hbar^2}{2m^*}Q^2 - \mu\right) - \frac{\hbar^2\beta}{2m^*}\mathbf{P}\cdot\mathbf{Q} \tag{39}$$

$$= X - \xi\mathbf{P}\cdot\mathbf{Q}. \tag{40}$$

Similarly,

$$\beta\left(\varepsilon(\mathbf{p}_2) - \mu\right) = X + \xi\mathbf{P}\cdot\mathbf{Q}. \tag{41}$$

[1] X can possibly be interpretted as: *average deviation from background chemical potential*

Hence,

$$f_0(\mathbf{p}_1) = \frac{1}{\exp[\beta(\varepsilon(\mathbf{p}_1) - \mu)]} = \frac{1}{\exp[X - \xi \mathbf{P} \cdot \mathbf{Q}]}. \quad (42)$$

$$f_0(\mathbf{p}_2) = \frac{1}{\exp[\beta(\varepsilon(\mathbf{p}_1) - \mu)]} = \frac{1}{\exp[X + \xi \mathbf{P} \cdot \mathbf{Q}]} \quad (43)$$

Recall that $F_{121'2'} = f_0(\mathbf{p}_1)f_0(\mathbf{p}_2)(1-f_0(\mathbf{p}_1'))(1-f_0(\mathbf{p}_2'))$

$$f_0(\mathbf{p}_1)f_0(\mathbf{p}_2) = \frac{1}{\exp[X-\xi\mathrm{P}\cdot\mathrm{Q}]\exp[X+\xi\mathrm{P}\cdot\mathrm{Q}]} \tag{44}$$

$$= \frac{1}{\exp[2X]+\exp[X](\exp[\xi\mathrm{P}\cdot\mathrm{Q}]+\exp[-\xi\mathrm{P}\cdot\mathrm{Q}])+1} \tag{45}$$

$$= \frac{1}{\exp[2X]+2\exp[X]\cosh(\xi\mathrm{P}\cdot\mathrm{Q})+1} \tag{46}$$

$$= \frac{1}{2\left(\cosh X+\cosh(\xi\mathrm{P}\cdot\mathrm{Q})\right)} \tag{47}$$

By symmetry,

$$(1-f_0(\mathbf{p}_1'))(1-f_0(\mathbf{p}_2')) = \frac{1}{2\left(\cosh X+\cosh(\xi\mathrm{P}'\cdot\mathrm{Q}')\right)} \tag{48}$$

$$F_{121'2'} = \frac{1}{4[\cosh X+\cosh(\xi\mathrm{P}\cdot\mathrm{Q})][\cosh X+\cosh(\xi\mathrm{P}\cdot\mathrm{Q})]} \tag{49}$$

Expression of W - the collision rate

$$W(\mathbf{p}_1', \mathbf{p}_2' | \mathbf{p}_1, \mathbf{p}_2) = \frac{2\pi}{\hbar} |V|^2 \times (2\pi)^2 \times \delta(\mathbf{p}_1 + \mathbf{p}_2 - \mathbf{p}_1' - \mathbf{p}_2') \delta\left(\varepsilon(\mathbf{p}_1) + \varepsilon(\mathbf{p}_2) - \varepsilon(\mathbf{p}_1') - \varepsilon(\mathbf{p}_2')\right) \quad 50$$

$$\hat{\mathcal{J}}[f(t, \mathbf{p}_1)] = -\frac{1}{(2\pi)^6} \int \dots \int \mathrm{d}\mathbf{p}_2 \, \mathrm{d}\mathbf{p}_1' \, \mathrm{d}\mathbf{p}_2' \times \frac{2\pi}{\hbar} |V|^2 \times (2\pi)^2 \times \delta(\mathbf{p}_1 + \mathbf{p}_2 - \mathbf{p}_1' - \mathbf{p}_2') \delta\left(\varepsilon(\mathbf{p}_1) + \varepsilon(\mathbf{p}_2) - \varepsilon(\mathbf{p}_1') - \varepsilon(\mathbf{p}_2')\right) \times F_{121'2'} \left[\psi(\mathbf{p}_1) + \psi(\mathbf{p}_2) - \psi(\mathbf{p}_1') - \psi(\mathbf{p}_2')\right] \quad 51$$

Change of coordinates: $\mathbf{P} = \mathbf{p}_1 + \mathbf{p}_2$, $\mathbf{Q} = \dfrac{1}{2}(\mathbf{p}_1 - \mathbf{p}_2)$.

$$
\begin{aligned}
\hat{\mathcal{J}}[f(t,\mathbf{p}_1)] = & -\frac{1}{(2\pi)^3\hbar}\int\dots\int \mathrm{d}\mathbf{p}_2\,\mathrm{d}\mathbf{P}'\,\mathrm{d}\mathbf{Q}'\,\delta(\mathbf{P}-\mathbf{P}')\,\delta\left(\frac{\hbar^2}{2m^*}(p_1^2+p_2^2-p_1'^2-p_2'^2)\right) \\
& \times |V|^2 F_{121'2'}\left[\psi(\mathbf{p}_1)+\psi(\mathbf{p}_2)-\psi(\mathbf{p}_1')-\psi(\mathbf{p}_2')\right] \\
= & -\frac{1}{(2\pi)^3\hbar}\int\dots\int \mathrm{d}\mathbf{p}_2\,\mathrm{d}\mathbf{Q}'\,\delta\left(\frac{\hbar^2}{4m^*}(4Q^2-4Q'^2)\right) \\
& \times |V|^2 F_{121'2'}\left[\psi(\mathbf{p}_1)+\psi(\mathbf{p}_2)-\psi(\mathbf{p}_1')-\psi(\mathbf{p}_2')\right] \\
= & -\frac{1}{(2\pi)^3\hbar}\int\dots\int \mathrm{d}\mathbf{p}_2 Q'\,\mathrm{d}Q'\,\mathrm{d}\Omega\,\frac{m^*}{\hbar^2}\delta(Q^2-Q'^2) \\
& \times |V|^2 F_{121'2'}\left[\psi(\mathbf{p}_1)+\psi(\mathbf{p}_2)-\psi(\mathbf{p}_1')-\psi(\mathbf{p}_2')\right] \\
= & -\frac{m^*}{4\pi\hbar^3}\int\dots\int\frac{\mathrm{d}\mathbf{p}_2}{(2\pi)^2}\int \mathrm{d}\Omega |V|^2 F_{121'2'}\left[\psi(\mathbf{p}_1)+\psi(\mathbf{p}_2)-\psi(\mathbf{p}_1')-\psi(\mathbf{p}_2')\right]
\end{aligned}
$$

2.2 Eigenvalue problem

Reducing the Boltzmann equation

$$\left(\frac{\partial}{\partial t} + \mathbf{v}\cdot\frac{\partial}{\partial \mathbf{r}} + \mathbf{F}\cdot\frac{\partial}{\partial \hbar\mathbf{p}}\right) f(t,\mathbf{r},\mathbf{p}) = \hat{\mathcal{J}}[f(t,\mathbf{r},\mathbf{p})] \tag{52}$$

- No external force: $\mathbf{F} = \mathbf{0}$,
- No thermal gradient: $\dfrac{\partial f}{\partial \mathbf{r}} = 0$.

The equation reduces to

$$\frac{\partial f}{\partial t} = \hat{\mathcal{J}}[f] \tag{53}$$

Recall that

- $f(t,\mathbf{p}) = f_0(\mathbf{p}) + \delta f(t,\mathbf{p})$
- $\delta f(t,\mathbf{p}) = f_0(1-f_0)\psi(t,\mathbf{p})$
- $\dfrac{\partial f_0}{\partial t} = 0, \quad \hat{\mathcal{J}}[f_0] = 0$

Substituting,

$$\frac{\partial}{\partial t}(f_0(\mathbf{p}) + \delta f(t,\mathbf{p})) = \hat{\mathcal{J}}[f_0(\mathbf{p}) + \delta f(t,\mathbf{p})] \tag{54}$$

$$\frac{\partial}{\partial t} f_0(1-f_0)\psi(t,\mathbf{p}) = \hat{\mathcal{J}}[f_0(1-f_0)\psi(t,\mathbf{p})] \tag{55}$$

$$\frac{\partial \psi(t,\mathbf{p})}{\partial t} = \hat{\mathcal{J}}[\psi(t,\mathbf{p})] \tag{56}$$

Proposed ansatz $\psi(t, \mathbf{p}) = \mathrm{e}^{-\gamma t}\phi(\mathbf{p})$ (Chemical potential decays with time)

So, the equation reduces to an eigenvalue problem:

$$\hat{\mathcal{J}}[\phi(\mathbf{p})] = -\gamma\phi(\mathbf{p})$$

If *perturbation is smooth*, it can be *expanded in the circular harmonics basis*:

$$\mathrm{e}^{im\theta}, \quad m \in \mathbb{Z} \tag{58}$$

For a test mode m,

$$\hat{\mathcal{J}}[\mathrm{e}^{im\theta}] = -\gamma_m \mathrm{e}^{im\theta} \tag{59}$$

Our goal is to study the eigenvalues γ_m as a function of m and temperature T.

3 Odd-Even effect

Trivial solution

$\gamma_m = 0$ is associated with **conserved quantities**.

- $\gamma_0 = 0$
  1. $\phi_0 = 1$: particle number
  2. $\phi_0 = \varepsilon(\mathbf{p})$: energy
- $\gamma_1 = 0$
  1. $\phi_1 = \mathrm{e}^{i\theta}$: x-momentum
  2. $\phi_1 = \mathrm{e}^{-i\theta}$: y-momentum

Near the Fermi surface,

$$\varepsilon(\mathbf{p}_1) \approx \varepsilon(\mathbf{p}_2) \approx \varepsilon(\mathbf{p}_1') \approx \varepsilon(\mathbf{p}_2') \approx \varepsilon_F. \quad (60)$$

Thus, **conservation of energy** implies **conservation of momentum magnitude**:

$$p_1 + p_2 = p_1' + p_2'. \quad (61)$$

Three types of collisions that satisfy this constraint:

- Head-on collisions: $\mathbf{p}_1 = -\mathbf{p}_2, \mathbf{p}_1' = -\mathbf{p}_2'$
- Forward exchange: $\mathbf{p}_1' = \mathbf{p}_2, \mathbf{p}_2' = \mathbf{p}_1$
- No collision: $\mathbf{p}_1' = \mathbf{p}_1, \mathbf{p}_2' = \mathbf{p}_2$

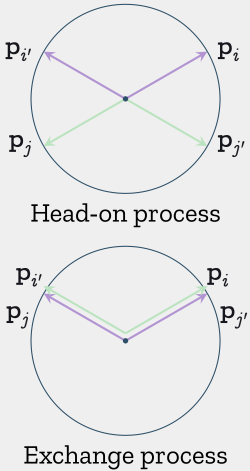

Figure 4: Relevant collision types

No collisions: Has no contribution to the collision integral.

Forward exchange & No collisions:

- $\mathbf{p}_1 = \mathbf{p}_1', \mathbf{p}_2 = \mathbf{p}_2'$.
- $\psi(\mathbf{p}_1) + \psi(\mathbf{p}_2) - \psi(\mathbf{p}_1') - \psi(\mathbf{p}_2') = 0$[2]
- $\therefore$ These collisions has no contribution to the collision integral.

Only **head-on collision** contribute to the collision integral.

[2] Recall that this is a multiplier in the collision integral

Head on collision

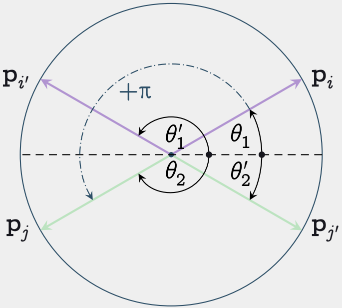

Figure 5: HEAD-ON COLLISION ANGLE

$$\theta_2 = \theta_1 + \pi, \quad \text{and} \quad \theta_2' = \theta_1' + \pi. \tag{62}$$

Substituting test mode m, and recalling $e^{im(\theta+\pi)} = (-1)^m e^{im\theta}$, then

$$\psi(\mathbf{p}_1) + \psi(\mathbf{p}_2) - \psi(\mathbf{p}_1') - \psi(\mathbf{p}_2') = (1 + (-1)^m)e^{im\theta_1} - (1 + (-1)^m)e^{im\theta_{1'}}. \tag{63}$$

***m* is odd:** the term vanishes; thus the collision integral vanishes.

***m* is even:** the term does not vanish.

Finding the Even Decay Rate : Overview

- Use **auxiliary variable** $\delta_1 = X - \xi \mathrm{P} \cdot \mathrm{Q}$ and $\delta_2 = X + \xi \mathrm{P} \cdot \mathrm{Q}$, together with $\delta_1 = 0$, and $\theta_{12} = \theta_2 - \theta_1$.
- Collision integral over $\mathbf{p}_2$ and $\boldsymbol{\Omega}$. Decompose $\mathbf{p}_2$ into **magnitude** p_2 and **angle** θ_2.
- **Change integration variable** from p_2 to δ_2 and θ_2 to $\theta_{12} = \theta_2 - \theta_1$.
- **Omega integral**: Approximate far from $\theta_{12} = 0, \pi$, reduces to standard hyperbolic integral. Gains a T factor.
- δ_2 **integral**: Standard numeric integral. Gains another T factor from change of variables.
- θ_{12} **integral**: Seemingly diverges, but cut off by Ω integral. Gains a $\log(T)$ factor.

Finding the Even Decay Rate : Auxiliary Variables

Recall $X - \xi \mathrm{P} \cdot \mathrm{Q} = \beta(\varepsilon(\mathbf{p}_1) - \mu)$, $X + \xi \mathrm{P} \cdot \mathrm{Q} = \beta(\varepsilon(\mathbf{p}_2) - \mu)$. Define

$$\delta_1 = X - \xi \mathrm{P} \cdot \mathrm{Q}, \quad \text{and} \quad \delta_2 = X + \xi \mathrm{P} \cdot \mathrm{Q}. \tag{64}$$

$$\Longrightarrow X = \frac{\delta_1 + \delta_2}{2}, \quad \text{and} \quad \xi \mathrm{P} \cdot \mathrm{Q} = \frac{\delta_2 - \delta_1}{2} \tag{65}$$

Assuming $\varepsilon(\mathbf{p}_1) \approx \mu$, $\delta_1 \approx 0$, so $X \approx \xi \mathrm{P} \cdot \mathrm{Q} \approx \dfrac{\delta_2}{2}$.

Finding the Even Decay Rate : Auxiliary Variables

Define $\theta_{12} = \theta_2 - \theta_1$. Then

$$P = 2p_F|\cos(\theta_{12}/2)|, \quad \text{and} \quad Q = p_F|\sin(\theta_{12}/2)|. \tag{66}$$

$$\implies \xi\mathbf{P}\cdot\mathbf{Q}' = \xi PQ\sin\Omega = \xi p_F^2|\sin(\theta_{12})|\sin\Omega. \tag{67}$$

(68)

Let $\Lambda = \varepsilon_F\beta = \xi p_F^2$. Finally, $\xi\mathbf{P}\cdot\mathbf{Q}' = \Lambda|\sin(\theta_{12})|\sin\Omega$.

Finding the Even Decay Rate : Integration Variables

The collision integral has the form (note the first (...) contains constants, the second contains dimensionless inteegration quantities like ϕ).

$$(\ldots)\int\ldots\int \mathrm{d}\mathbf{p}_2\,\mathrm{d}\Omega F_{121'2'}(\ldots) \tag{69}$$

Using $\mathrm{d}\mathbf{p}_2 = p_2\,\mathrm{d}p_2\,\mathrm{d}\theta_2, \mathrm{d}\theta_2 = \mathrm{d}\theta_{12}$, and $\mathrm{d}\delta_2 = \beta\,\mathrm{d}\varepsilon(p_2) = \beta\frac{\hbar^2 p_2}{m^*}$, we obtain the form

$$(\ldots)\int\ldots\int \mathrm{d}\delta_2\,\mathrm{d}\theta_{12}\,\mathrm{d}\Omega F_{121'2'}\cdot(\ldots) \tag{70}$$

with constants absorbed into the first (...) (restored later on). Recall $\beta = \frac{1}{k_B T}$. **This is our first T factor**.

Finding the Even Decay Rate : $F_{121'2'}$

$$F_{121'2'} = \frac{1}{4[\cosh X + \cosh \xi \mathbf{P}\cdot\mathbf{Q}][\cosh X + \cosh \xi \mathbf{P}\cdot\mathbf{Q}']} \tag{71}$$

$$= \frac{1}{4[\cosh(\delta_1+\delta_2)/2 + \cosh(\delta_2-\delta_1)/2][\cosh(\delta_1+\delta_2)/2 + \cosh(\Lambda|\sin(\theta_{12})|\sin\Omega)]} \tag{72}$$

$$= \frac{1}{4[\cosh\delta_2/2 + \cosh\delta_2/2][\cosh\delta_2/2 + \cosh(\Lambda|\sin(\theta_{12})|\sin\Omega)]} \tag{73}$$

$$= \frac{1}{8\cosh\delta_2/2[\cosh\delta_2/2 + \cosh(\Lambda|\sin(\theta_{12})|\sin\Omega)]}. \tag{74}$$

where we have used $\delta_1 = 0$.

Finding the Even Decay Rate : Omega Integral

Collecting terms that depend on Ω, we have

$$\int_0^{2\pi} \frac{\mathrm{d}\Omega}{\cosh \delta_2/2 + \cosh\left(\Lambda |\sin(\theta_{12})| \sin \Omega\right)}. \tag{75}$$

For θ_{12} not near 0 or π, approximate $\sin \Omega \approx u$, extend the integral to $(-\infty, \infty)$, and change integration variables to $v = \Lambda |\sin(\theta_{12})| u$. Then the integral becomes

$$= \int_{-\infty}^{\infty} \frac{\mathrm{d}\Omega}{\cosh \delta_2/2 + \cosh\left(\Lambda |\sin(\theta_{12})| \Omega\right)} = \frac{1}{\Lambda |\sin(\theta_{12})|} \int_{-\infty}^{\infty} \frac{\mathrm{d}v}{\cosh \delta_2/2 + \cosh v} \tag{76}$$

$$= \frac{2\delta_2}{\Lambda |\sin(\theta_{12})| \sinh \delta_2/2} \tag{77}$$

Where we have used the standard integral $\int_{-\infty}^{\infty} \frac{\mathrm{d}x}{\cosh a + \cosh x} = \frac{2a}{\sinh a}$. (**Lambda is our second T factor**).

Finding the Even Decay Rate : δ_{12} Integral

Collecting terms that depend on δ_2, we have

$$\frac{1}{8\cosh\delta_2/2}\,\frac{2\delta_2}{\Lambda\sin(\theta_{12})|\sinh\delta_2/2} = \frac{\delta_2}{2\Lambda\sin(\theta_{12})|\sinh\delta_2}. \qquad 78$$

Using the standard integral $\int_{-\infty}^{\infty}\frac{x}{\sinh x}\,\mathrm{d}x = \frac{\pi^2}{2}$, we have

$$\int_{-\infty}^{\infty}\frac{\delta_2}{2\Lambda\sin(\theta_{12})|\sinh\delta_2}\,\mathrm{d}\delta_2 = \frac{\pi^2}{4\Lambda|\sin(\theta_{12})|}. \qquad 79$$

Finding the Even Decay Rate : θ_{12} Integral

Collecting terms that depend on θ_{12}, we have

$$\int_0^{2\pi} \frac{\mathrm{d}\theta_{12}}{\Lambda|\sin(\theta_{12})|}. \tag{80}$$

The integral is divergent at 0 and π. However, it is cut off by the Ω integral. The cutoff occurs when $\Lambda|\sin(\theta_{12})| \sim 1 \implies |\theta_{12}| \sim T/T_F$. Elsewhere, the antiderivative equals $\ln(\tan(\theta_{12}/2)) + C$. So, for θ_{12} at least T/T_F from 0 or π, we have

$$= \int_{T/T_F}^{\pi - T/T_F} \frac{\mathrm{d}\theta_{12}}{\Lambda|\sin(\theta_{12})|}. \tag{81}$$

$$= \frac{1}{\Lambda}[\ln(\tan((\pi - T/T_F)/2)) - \ln(\tan(T/T_F/2))] \tag{82}$$

$$= \frac{1}{\Lambda}[-2\ln(\tan(T/T_F/2))] \tag{83}$$

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Finding the Even Decay Rate : Final Result

Collecting all the T factors, we have

- One from β in the change of variables from p_2 to δ_2.
- One from the Λ from Ω integral.
- One log factor from the θ_{12} integral.

Combining constants obtained during the change of integration variable, we obtain

Collision Integral Approximation to Zero-th order

$$-\gamma_m \phi = \hat{\mathcal{J}}[\phi] = -\frac{k_B \pi^2 m^{*2} |V|^2}{\hbar^4 (2\pi)^3} \frac{T^2}{T_F} \ln(\frac{T_F}{T}) \phi \tag{86}$$

Thus, we have $\gamma_m \propto T^2 \log T$. Note that this only works for small ϕ.

Strategy for the Odd Decay Rate

Since, to the zeroth order, the ϕ terms vanish, we need the second order perturbation (first order also vanishes). The strategy is to

- Develop eigenvalue perturbation around $\delta_T = T/T_F \ll 1$
- Expand the ϕ around 0 to obtain a power series in $\varepsilon(p_i)/\varepsilon_F = \delta_T u_i$, where $u_i = \varepsilon_i/T$.
- Integrate the power series against the Fermi functions.
- Use the eigenvalue perturbation theory to obtain the lowest order approximation of γ_m.

4 Further works

van Hove singularity

van Hove singularity: An **energy level** that has infinitely many available states.[a]

[a] Spike in the density of states

- More available states $\implies$ stronger pairing interactions (forming Cooper pairs)
- Superconductivity can happen

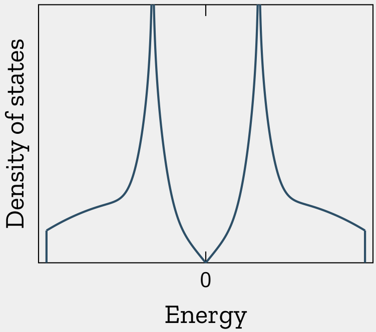

Figure 6: van-Hove singularity (not to scale, not real data)

If Fermi surface has the **energy that's close to the van Hove singularity**, superconductivity can occur more easily.

Questions:

- Does odd-even effect near van Hove singularity?
- What's the relaxation time?